

Word Embeddings

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1. Introduction

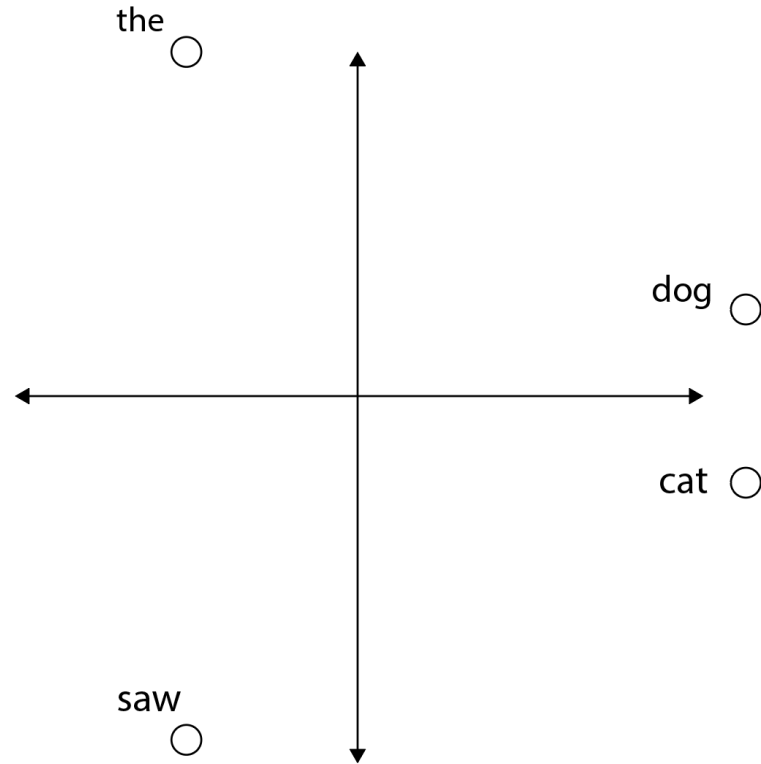
domestic dog clade: *familiaris*, as named by Linnaeus in 1758 and, dingo named by Meyer in 1793. Wozencraft included *hallstromi* (the New Guinea singing dog) as another name (junior synonym) for the dingo. Wozencraft referred to the mtDNA study as one of the guides informing his decision. Mammalogists have noted the inclusion of *familiaris* and dingo together under the domestic dog clade with some debating it. In 2019, a workshop hosted by the IUCN/Species Survival Commission's Canid Specialist Group considered the dingo and the New Guinea singing dog to be feral *Canis familiaris* and therefore did not assess them for the IUCN Red List of Threatened Species.

Evolution The Cretaceous-Paleogene extinction event occurred 65 million years ago and brought an end to the non-avian dinosaurs and the appearance of the first carnivorans. The name carnivoran is given to a member of the order Carnivora. Carnivorans possess a common arrangement of teeth called carnassials, in which the first lower molar and the last upper premolar possess blade-like enamel crowns that act similar to a pair of shears for cutting meat. This dental arrangement has been modified by adaptation over the past 60 million years for diets composed of meat, for crushing vegetation, or for the loss of the carnassial function altogether as in seals, sea lions, and walruses. Today, not all carnivorans are carnivores, such as the insect-eating aardwolf. The carnivoran ancestors of the dog-like caniforms and the cat-like feliforms began their separate evolutionary paths just after the end of the dinosaurs. The first members of the dog family Canidae appeared 40 million years ago, of which only its subfamily the Caninae survives today in the form of the wolf.

- Counting words / n-grams / PMI
 - Conceptually straightforward and computationally fast
 - Not very flexible
 - All words are treated as independent categories

1.3 Word embeddings: Toy example

“the dog saw the cat” →



1.4 Toy example: math

$$\text{dog} = \begin{bmatrix} 0.23 \\ -1.47 \\ 1.01 \\ 1.33 \\ \dots \\ -0.56 \end{bmatrix}, \text{cat} = \begin{bmatrix} 0.42 \\ -1.07 \\ 0.31 \\ 1.54 \\ \dots \\ 0.14 \end{bmatrix} \in \mathbb{R}^{100}$$

```
from probabilistic_word_embeddings.embeddings import  
Embedding
```

```
e = Embedding(saved_model_path="embedding.pkl")  
dog = e["dog"]  
cat = e["cat"]
```

1.6 Using word embeddings

- Distance / similarity
 - Cosine distance / similarity
 - How similar are words x, y
 - What is the most similar word to word x
- Analogies
 - king — man + woman $\approx$ queen
- And more

Cosine similarity

$$\text{cossim}(\rho_v, \rho_w) = \frac{\rho_v \cdot \rho_w}{\|\rho_v\| \|\rho_w\|}$$

Differences

$$\rho_v - \rho_w$$

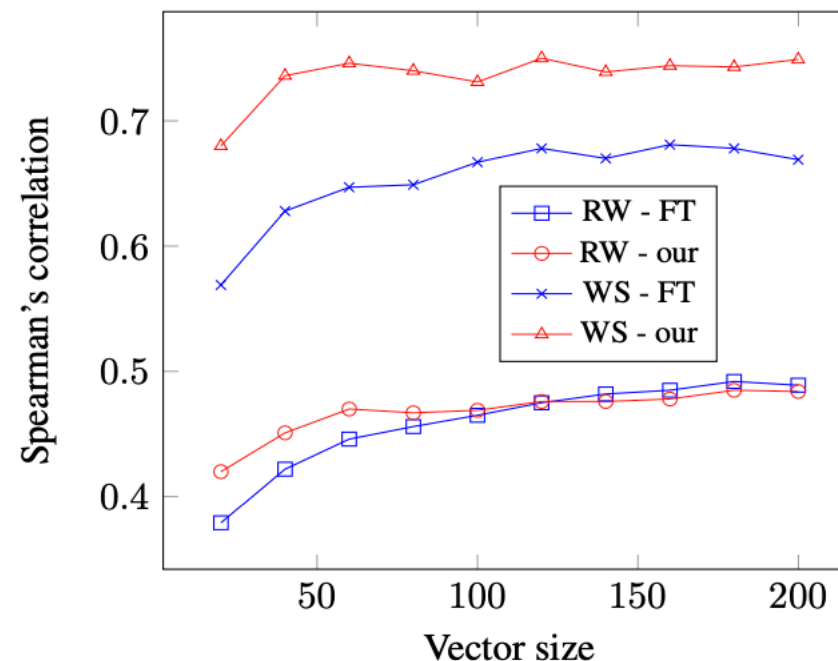


Figure 1: Spearman's rank correlation coefficient for RW-STANFORD (RW) and WS-353-ALL (WS) on the fastText model (FT) and our, with different vector size. Training is done on the corpus A of 50M tokens.

```
e1 = Embedding(saved_model_path="dict2vec_embedding.pkl")
e2 = Embedding(saved_model_path="fasttext_embedding.pkl")

similarity1 = cossim(e1["dog"], e1["cat"])
similarity2 = cossim(e2["dog"], e2["cat"])
```

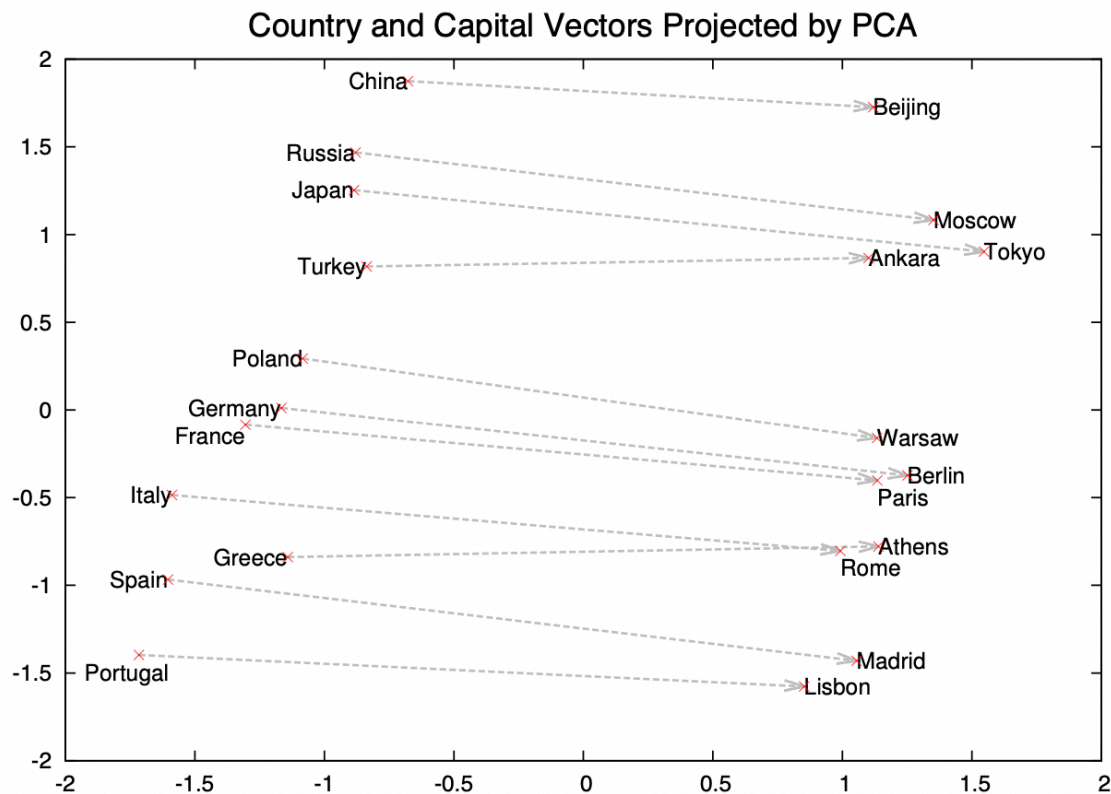


Figure 2: Two-dimensional PCA projection of the 1000-dimensional Skip-gram vectors of countries and their capital cities. The figure illustrates ability of the model to automatically organize concepts and learn implicitly the relationships between them, as during the training we did not provide any supervised information about what a capital city means.

```
diff1 = e["china"] - e["beijing"]  
diff2 = e["germany"] - e["berlin"]
```

1.12 Training embeddings vs pretrained

- For research, you can do two things
 - Use a pre-trained embedding
 - Gather data and train your own embedding
- It is crucial what data has been used for training
- Own embeddings can be more specific, but require work and good data

2. Training word embeddings

“you shall know a word by the company it keeps”

— Firth (1957)

2.2 Example

2. Training word embeddings

domestic **dog** clade: familiaris, as named by Linnaeus in 1758 and, dingo named by Meyer in 1793. Wozencraft included hallstromi (the New Guinea singing **dog**) as another name (junior synonym) for the dingo. Wozencraft referred to the mtDNA study as one of the guides informing his decision. Mammalogists have noted the inclusion of familiaris and dingo together under the domestic **dog** clade with some debating it. In 2019, a workshop hosted by the IUCN/Species Survival Commission's Canid Specialist Group considered the dingo and the New Guinea singing **dog** to be feral Canis familiaris and therefore did not assess them for the IUCN Red List of Threatened Species. Evolution The Cretaceous-Paleogene extinction event occurred 65 million years ago and brought an end to the non-avian dinosaurs and the appearance of the first carnivorans. The name carnivoran is given to a member of the order Carnivora. Carnivorans possess a common arrangement of teeth called carnassials, in which the first lower molar and the last upper premolar possess blade-like enamel crowns that act similar to a pair of shears for cutting meat. This dental arrangement has been modified by adaptation over the past 60 million years for diets composed of meat, for crushing vegetation, or for the loss of the carnassial function altogether as in seals, sea lions, and walruses. Today, not all carnivorans are carnivores, such as the insect-eating aardwolf. The carnivoran ancestors of the **dog**-like caniforms and the cat-like feliforms began their separate evolutionary paths just after the end of the dinosaurs. The first members of the **dog** family Canidae appeared 40 million years ago, of which only its subfamily

[...] as the oldest domesticated species, dogs' minds inevitably have been shaped by millennia of contact with humans [...]

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dog

{domesticated, species,
minds, inevitably}

2.6 Word2Vec – concept

2. Training word embeddings

- Continuous bag-of-words CBOW

Does the word x appear, given that its context window is

$y_1, y_2, \dots, y_{ws}$

- Skip-gram

Do words x and y appear in the same word window?

Mikolov et al. (2013a), Mikolov et al. (2013b)

[...] as the oldest domesticated
species, [???] minds inevitably have
been shaped by millenia of contact
with humans

⇒ Whole context window simulatenously!

[...] as the oldest **domesticated**
species **[???** minds inevitably have
been shaped by millenia of contact
with humans

[...] as the oldest domesticated
species [???] minds inevitably have
been shaped by millenia of contact
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⇒ One context word at the time!

- Each word is assigned a word vector ρ and a context vector α
 - $\rho, \alpha \in \mathbb{R}^D$, where D is the *dimensionality* of the embedding
- Conditional probabilities are functions of these vectors
 - $P(x \mid y_1, \dots, y_w) = f(\rho_x, \alpha_{y_1}, \dots, \alpha_{y_w})$

2.13 Skip-gram

2. Training word embeddings

Do words x and y appear in the same word window?

$$p(x \text{ and } y \text{ co-occur}) = \sigma(\rho_x^T \alpha_y)$$

where $\sigma(\cdot)$ is the logistic function.

2.14 Skip-gram, numerical example

2. Training word embeddings

$$\rho_{\text{dog}} = \begin{bmatrix} 0.23 \\ -1.47 \\ 1.01 \\ 1.33 \\ \dots \\ -0.56 \end{bmatrix}, \quad \alpha_{\text{cat}} = \begin{bmatrix} 0.42 \\ -1.07 \\ 0.31 \\ 1.54 \\ \dots \\ 0.14 \end{bmatrix}$$

Do the words *dog* and *cat* appear in the same word window?

$$\begin{aligned} p(\text{dog and cat co-occur}) &= \sigma \left([0.23 \quad -1.47 \quad 1.01 \quad 1.33 \quad \dots \quad -0.56] \begin{bmatrix} 0.42 \\ -1.07 \\ 0.31 \\ 1.54 \\ \dots \\ 0.14 \end{bmatrix} \right) \\ &= \sigma(6.2325) = 0.998 \end{aligned}$$

$$\rho_{\text{dog}} = \begin{bmatrix} 0.23 \\ -1.47 \\ 1.01 \\ 1.33 \\ \dots \\ -0.56 \end{bmatrix}, \quad \alpha_{\text{university}} = \begin{bmatrix} -2.03 \\ 0.07 \\ 1.01 \\ -0.34 \\ \dots \\ 0.89 \end{bmatrix}$$

Do the words *dog* and *university* appear in the same word window?

$$\begin{aligned} p(\text{dog and university co-occur}) &= \sigma \left([0.23 \quad -1.47 \quad 1.01 \quad \dots \quad -0.56] \begin{bmatrix} -2.03 \\ 0.07 \\ 1.01 \\ -0.34 \\ \dots \\ 0.89 \end{bmatrix} \right) \\ &= \sigma(-0.13) = 0.4675 \end{aligned}$$

2.18 Continuous bag-of-words

2. Training word embeddings

Do words x and y appear in the same word window?

$$p(x \text{ occurs given } C = y_1, \dots, y_{ws}) = \sigma \left(\rho_x^T \sum_{y \in C} \alpha_y \right)$$

where C is the context window, and $\sigma(\cdot)$ is the logistic function.

2.19 Negative sampling

2. Training word embeddings

- In the data, we can easily find instances where words do co-occur
 - “the dog saw the cat”
 - Eg. (dog,saw), (dog,cat) and (the, cat) co-occur
- No direct examples of words not co-occurring
- We need negative examples, otherwise all vectors are the same
- Solution: pick at random

2.20 Negative sampling

2. Training word embeddings

- For each word x in the data
 - Sample $n_s \in \mathbb{N}$ negative samples y'_1, y'_{n_s} from the empirical distribution of word types
 - Add a term $1 - p(x \text{ and } y' \text{ co-occur})$ in the likelihood

2.21 Negative sampling

Say we only have the data point *dog co-occurs with cat*. We draw a negative sample *university* from the empirical distribution of words in the data. The likelihood is

$$\begin{aligned} p(\text{data}) &= \underbrace{p(\text{dog and cat co-occur})}_{\text{positive sample}} \underbrace{(1 - p(\text{dog and university co-occur}))}_{\text{negative sample}} \\ &= 0.998 \cdot (1 - 0.4675) = 0.531435 \end{aligned}$$

For a real dataset, the log likelihood is

$$\log p(\mathcal{D} \mid w, c) = \sum_{i=1}^N \left(\underbrace{\sum_{y \in C_i} \log \sigma(\rho_{x_i}^T \alpha_y)}_{\text{positive samples}} + \underbrace{\sum_{y \in C_{ns}} \log(1 - \sigma(\rho_{x_i}^T \alpha_y))}_{\text{negative samples}} \right)$$

where $\mathcal{D} = (x_1, \dots, x_N)$ is the dataset indexed by $i \in \{1, \dots, N\}$, C_i is the context window at i and C_{ns} is a randomized context window generated from the empirical distribution of words.

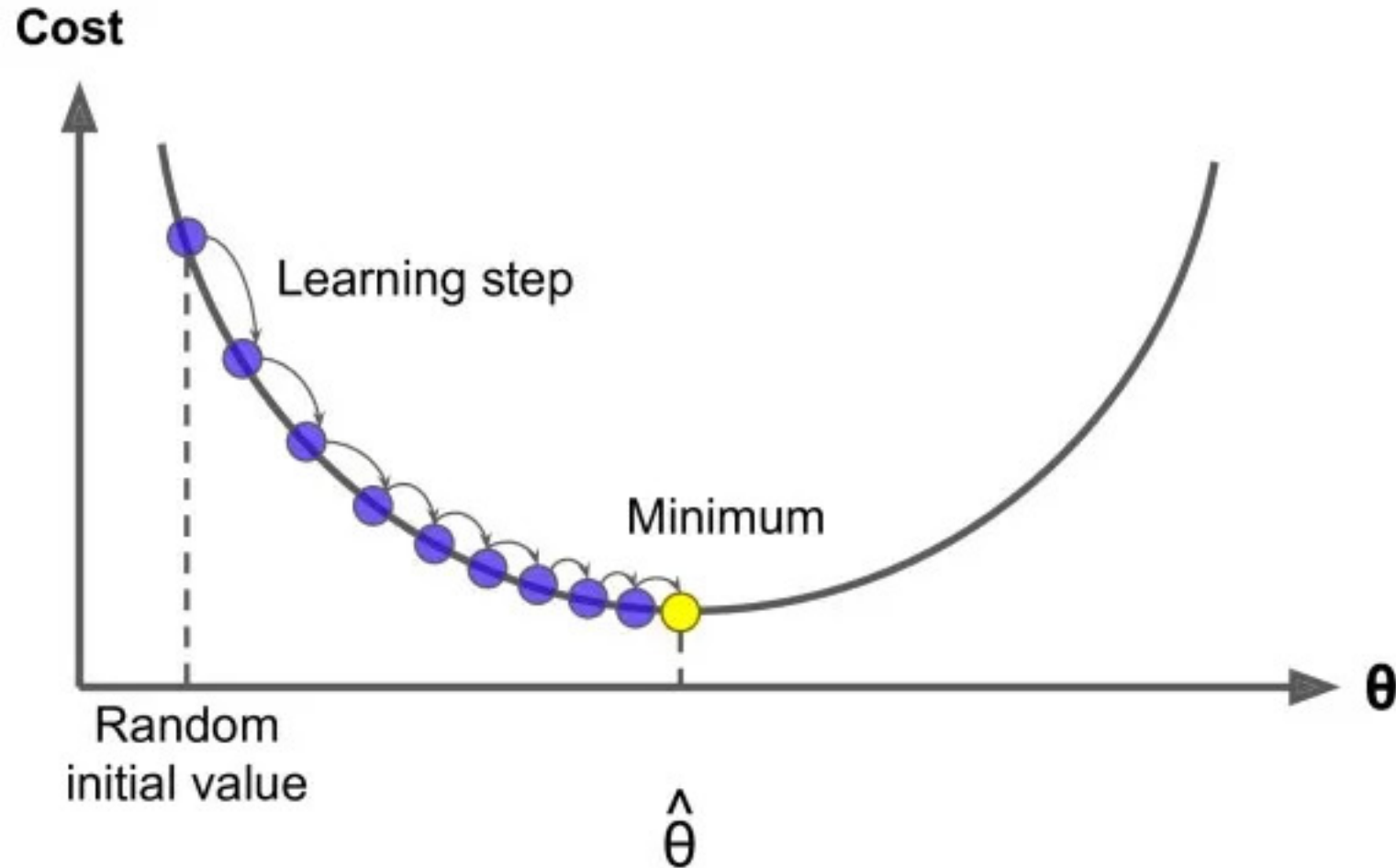
2.23 Gradient descent

2. Training word embeddings

- We want to minimize the loss (negative log likelihood)
- Gradient descent is used to iteratively do this
 - Start at random
 - At each step, see what direction improves the loss
 - Move to that direction
 - Repeat

2.24 Gradient descent

2. Training word embeddings



3. Different word embeddings

3.1 Probabilistic word embeddings

3. Different word embeddings

- Word2Vec as a probabilistic language model
 - Same likelihood
 - Add a prior on the parameters, eg. $\rho_x \sim \mathcal{N}(0, \lambda_0 I)$ for all word types $x \in W$
- Estimate via *maximum a posteriori* estimation, or *variational inference*, or *Gibbs sampling*

3.2 Bernoulli embeddings¹

3. Different word embeddings

$$\log p(w, c \mid \mathcal{D}) = \log p(\mathcal{D} \mid w, c) + \sum_{x \in W} \lambda_1 \|\rho_{x,t}\|^2 + \sum_{x \in W} \lambda_0 \|\alpha_x\|^2$$

Rudolph et al. (2016)

3.3 Probabilistic word embeddings

3. Different word embeddings

- Regularize models for data with natural divisions to subsets, eg. different time periods

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- Incorporate additional information about the words, eg. dictionary information

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- Regularize models for data with natural divisions to subsets, eg. different time periods
- Incorporate additional information about the words, eg. dictionary information
- Enforce properties of the embedding, eg. a “sentiment dimension”

3.6 Dynamic word embeddings

3. Different word embeddings

- Different set of word vectors for each time period $t \in \{1, \dots, T\}$
- For example, a separate vector for “dog” for each year
 - $\rho_{\text{dog},2012}, \dots, \rho_{\text{dog},2024}$
- You can calculate all the measurements for each year, or between years
 - Distance between $\rho_{\text{dog},2012}$ and $\rho_{\text{dog},2013}$
 - How does the similarity of $\rho_{\text{dog},t}$ and $\rho_{\text{cat},t}$ develop over time t ?

3.7 Dynamic word embeddings

3. Different word embeddings

- Random walk prior over the word vectors

$$\rho_{x,1} \sim \mathcal{N}(0, \lambda_0 I)$$

$$\rho_{x,t} \sim \mathcal{N}(\rho_{x,t-1}, \lambda_1 I) \quad \text{for } t \in \{2, \dots, T\}$$

- Spherical prior on context vectors

$$\alpha_x \sim \mathcal{N}(0, \lambda_0 I)$$

Bamler & Mandt (2017), Rudolph & Blei (2018)

3.8 Dynamic word embeddings

3. Different word embeddings

$$\begin{aligned}\log p(w, c \mid \mathcal{D}) &= \log p(\mathcal{D} \mid w, c) \\ &+ \sum_{x \in W} \sum_{t=1}^T \lambda_1 \|\rho_{x,t+1} - \rho_{x,t}\|^2 + \sum_{x \in W} \sum_{t=1}^T \lambda_0 \|\rho_{x,t}\|^2 \\ &+ \sum_{x \in W} \lambda_0 \|\alpha_x\|^2\end{aligned}$$

3.9 Informative priors

3. Different word embeddings

- Little data but also prior knowledge about the words
 - Still good embedding quality
- If we are interested in countries, we can add a prior between the different word forms
 - finland $\sim$ finnish $\sim$ finns $\sim$ finlands'
- Also other prior sources
 - Wordnet, synonyms

3.10 Graph priors

3. Different word embeddings

- Connections between words via graph priors
 - “These words are similar”
- Implemented in the pwe module
- Use for the dynamic prior in the Lab

3.11 Code example

3. Different word embeddings

```
from probabilistic_word_embeddings.embeddings import  
LaplacianEmbedding  
import networkx  
  
g = networkx.Graph()  
g.add_edge("finland", "finns")  
  
e = LaplacianEmbedding(vocabulary, graph=g)
```

3.12 Code example

3. Different word embeddings

```
from probabilistic_word_embeddings.embeddings import  
LaplacianEmbedding  
import networkx  
  
g = networkx.Graph()  
g.add_edge("dog_2012", "dog_2013")  
  
e = LaplacianEmbedding(vocabulary, graph=g)
```

4. Uncertainty estimation

4.1 Uncertainty estimation

4. Uncertainty estimation

- Many applications want to draw conclusions about the data
- Random noise vs real results?
- Methods to estimate embedding uncertainty
 - Probabilistic word embeddings (posterior sampling)
 - Bootstrapping
 - Frequentist CIs conditional on context vectors

4.2 Posterior Sampling of WEs

- Use a Gibbs sampler for estimation
 - Generate N posterior samples
 - Each sample is slightly different due to parameter uncertainty
- Calculate quantity of interest for all N word embeddings

Yrjänäinen et al. (2025)

4.3 Posterior Sampling of WEs

4. Uncertainty estimation

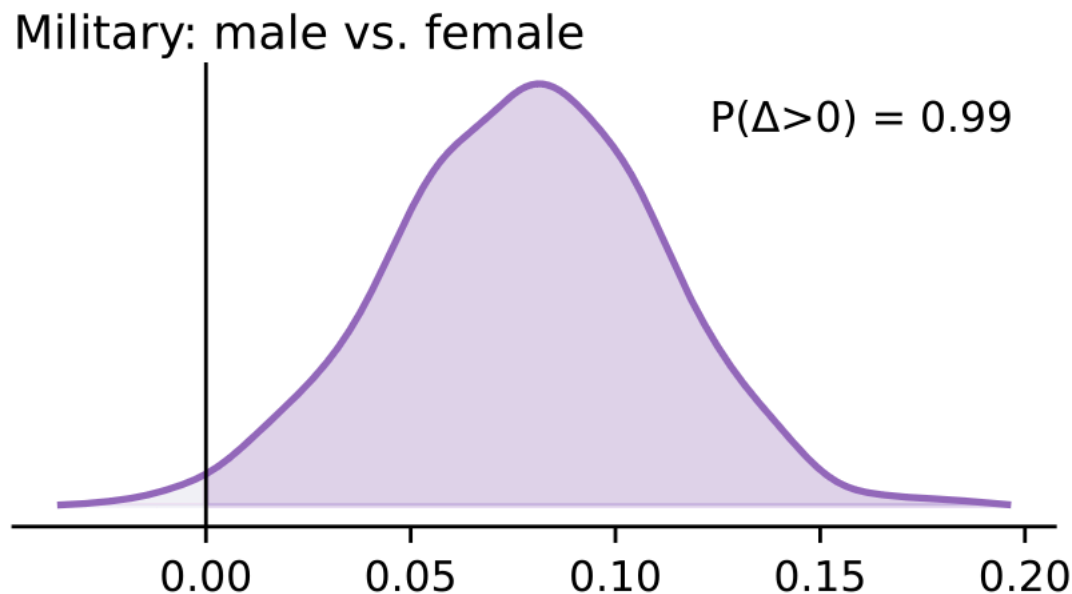


Figure 7: Posterior distribution of the association score for the US Congress data, computed from posterior samples of the Gibbs sampler ($N = 10,000,000$, $K = 100$).

4.4 Posterior Sampling of WEs

4. Uncertainty estimation

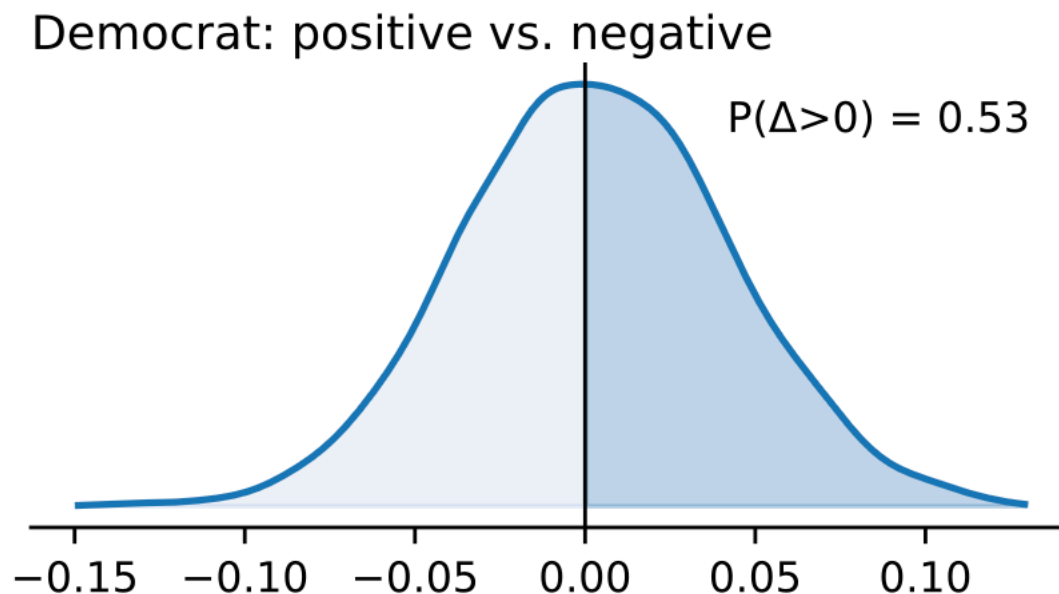


Figure 8: Posterior distribution of the association score for the US Congress data, computed from posterior samples of the Gibbs sampler ($N = 10,000,000$, $K = 100$).

4.5 Bootstrapping word embeddings

4. Uncertainty estimation

- Split corpus into documents that can be resampled
- Generate N bootstrap resamples
- Train the word embedding separately on all N resamples
- Calculate quantity of interest for all N word embeddings

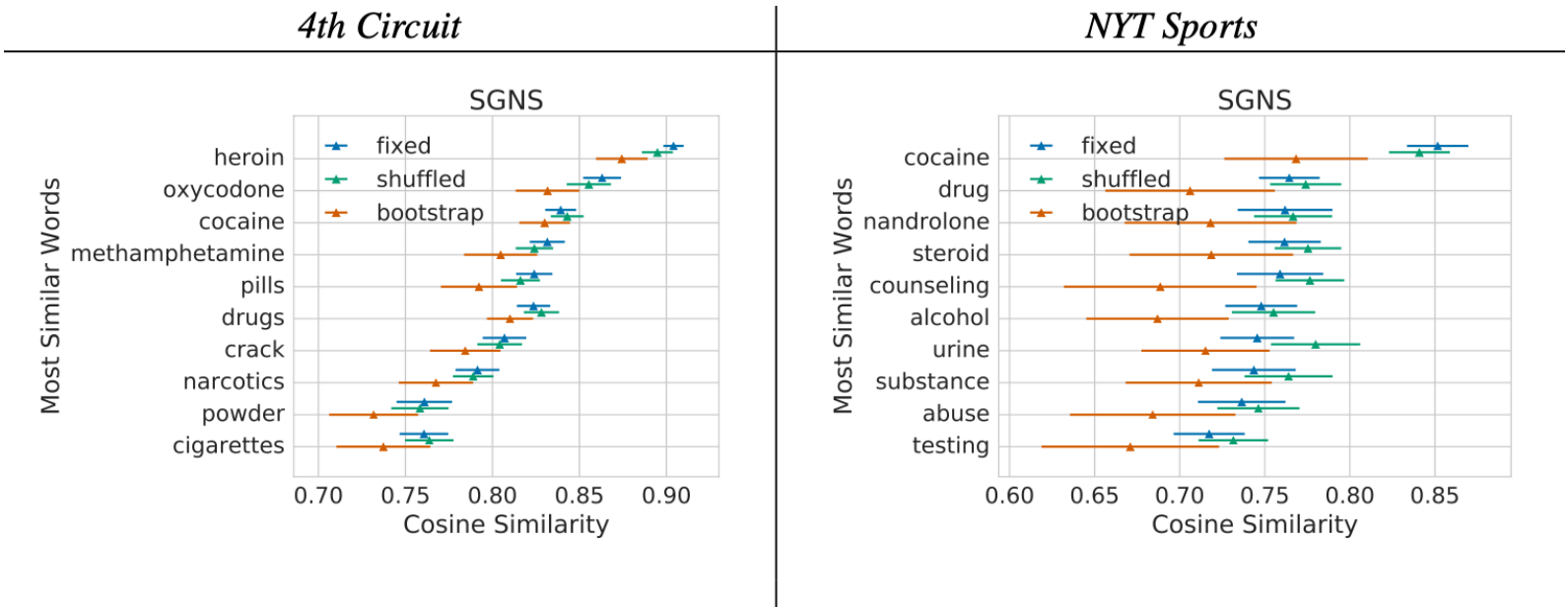


Figure 9: Most similar words to 'marijuana' in two corpora

4.7 Bootstrapping

4. Uncertainty estimation



4.8 Freq. Confidence Intervals

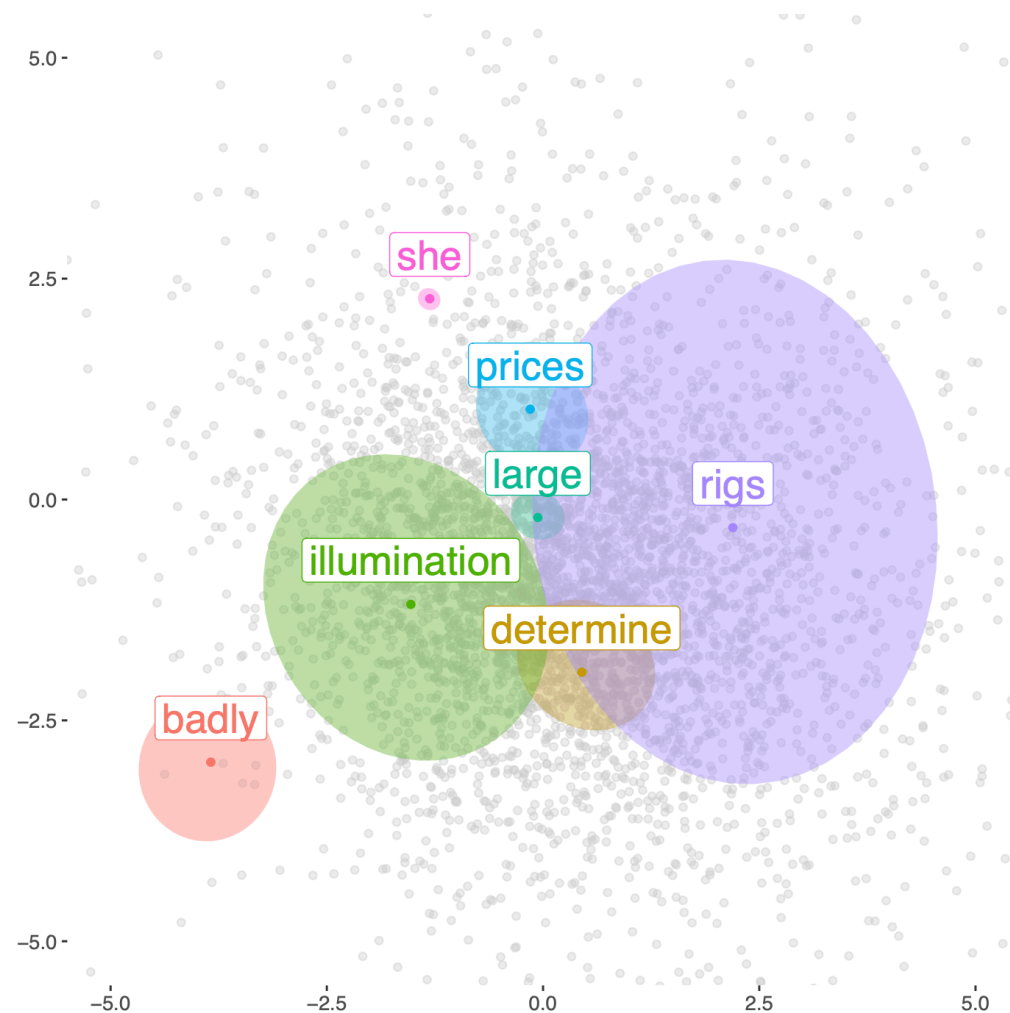
4. Uncertainty estimation

- Find embeddings via gradient descent
- Assume that context vectors have converged
 - The word vectors are then just a logistic regression
- Use confidence intervals for logistic regression

Vallebuena et al. (2024)

4.9 Freq. Confidence Intervals

4. Uncertainty estimation



4.10 Summary

- Word2vec yields vector representations of words that can be used to analyse words
 - Similarity, aspects, analogies
- Probabilistic embeddings enable the incorporation of prior knowledge and uncertainty estimation
 - Dynamic or grouped embeddings
 - Make estimation possible on smaller data via priors

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